AUTOMORPHIC AND GEOMETRIC REFORMULATIONS OF THE ERDŐS–STRAUS EQUATION, AND A DIVISOR-CLASS BARRIER

ANONYMOUS

Abstract. For a prime p we collect exact reformulations of the Erdős–Straus equation $4/p = 1/x + 1/y + 1/z$ in automorphic and arithmetic-geometric language. Type I solutions are $\mathrm{SL}_2(\mathbb{Z})$ matrices with a prescribed entry, equivalently Heegner points of discriminant $-4d$ in a thin wedge, whose discriminant-$(-4dp^2)$ forms are T_p-Hecke translates. Type II solutions are Heegner points of discriminant $-4dp$ lying on an exact shell around a cuspidal geodesic. Further reformulations use orders in the quaternion algebra $B_{p,\infty}$, the algebra $(-d,-p)_{\mathbb{Q}}$, a CM torus, and an isotropic ternary form. The Bright–Loughran Brauer class is a genus character; this is folklore and goes back to Yamamoto. All these encodings reduce to one statement, which we prove: once two of the Elsholtz–Tao coordinates are fixed, the solutions are either a finite union of congruence classes of p, or the divisors of one explicit integer $N_\theta(p)$ in one residue class. Every representative-independent invariant of the automorphic objects is congruence or genus data. We then record negative results, each with an exact scope. Fixed ternary embeddings reduce to Mordell coverings. Any detector must work at main term $p^{o(1)}$. Thickening a target is the same as averaging over n. Local, genus and spinor solubility hold even at odd squares. Under Schinzel's Hypothesis H, every argument that certifies the equation through a fixed finite set of slices fails infinitely often. We give the quantitative gaps to known equidistribution theorems, list the escape hatches that remain open, and report numerical evidence: $f_I(p) \geq 1$ and $f_{II}(p) \geq 1$ for all primes $p \leq 10^8$. The Erdős–Straus conjecture is *not* proved here. The negative statements concern the routes examined and known technology. They are not impossibility theorems for the conjecture.

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1. Introduction

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \qquad x, y, z \in \mathbb{N}, \tag{1.1}$$

is soluble. It suffices to treat primes $n = p$, and by Mordell's identities only primes in six classes modulo 840 (the squares) remain open. ES has been verified to very large bounds and holds for almost all n in a strong quantitative sense, but it is open for an individual large prime.

Why automorphic and geometric input. In earlier internal work of this campaign ([PO], and Theorems M and C of [PS]) we proved that *congruence-covering* arguments cannot settle ES pointwise. More precisely, under Schinzel's Hypothesis H, every correct bounded witness-producing procedure built from ring operations, floor division, sign tests and factorisations of the integers it meets fails for infinitely many primes $p \equiv 1 \pmod{24}$ [PS, Thm C]. This is the procedure-level form of the classical odd-square obstructions (Yamamoto, Mordell–Schinzel, Elsholtz–Tao [ET, Prop. 1.6], Bright–Loughran [BL, Cor. 1.4]). It leaves room for input that is *not* congruence data: spectral, automorphic or arithmetic-geometric information about a fixed prime p. This note collects what such input can see.

What this note contains.

(1) *Exact dictionaries* (§2, PROVED). Type I and Type II solutions are rewritten as Heegner points, SL_2 matrices, Hecke translates, shells around cuspidal geodesics, orders in definite quaternion algebras, divisors of an element of a CM field, and points on an isotropic ternary quadric. Several of these are folklore. We mark attributions explicitly; in particular, the identification of the Bright–Loughran Brauer class with a genus character is *not* new (Yamamoto 1965, via [BL, App. A]).
(2) *The divisor-class reduction* (§3, PROVED). Each encoding is a "slice" of the Elsholtz–Tao variety. Fixing two coordinates leaves either a congruence condition on p or the divisors of one integer $N_\theta(p)$ in one residue class (Lemma 3.1, Theorem 3.2).
(3) *Negative results with exact scope* (§4): ternary embeddings, the scale constraint, thickening, local solubility at odd squares, geometry, and Hypothesis-H-conditional emptiness of single targets.
(4) *Quantitative gaps* to known equidistribution theorems (§5, Assessment), *open problems* (§6), and *numerical evidence* (§7).

What this note does not contain. ES is not proved here, conditionally or unconditionally. We know of no standard conjecture (GRH, GLH, Ramanujan, Selberg's eigenvalue conjecture, twisted Linnik–Selberg) that is known to imply ES. The literature survey behind this note [LA] found no prior pointwise result for ES, conditional or not. Every negative statement below concerns the specific encodings and mechanisms analysed, or "known technology". None is a theorem that ES cannot be proved by automorphic methods. Words like "barrier" are used in that mechanism-scoped sense; where they summarise more than a proved statement, they are labelled Assessment.

Labels. Each statement carries one of the labels PROVED (a complete proof here, or in the cited internal file, relative to explicitly cited literature), CONDITIONAL (proved assuming a named hypothesis), CERTIFIED (an exact finite computation), EVIDENCE (numerics), or ASSESSMENT (a judgement that is not a theorem). Proofs of short items are given in full. For longer ones we give a sketch and a pointer to the internal source file, which also records its internal review. All internal files have been reviewed internally only; this note has not been externally refereed.

Notation. $p \geq 5$ is prime, $\mathcal{L} = \log p$, and $\chi_D = \left(\frac{D}{\cdot}\right)$ is the Kronecker symbol. For a negative discriminant D, $\chi_D(-1) = -1$. We write $[A, B, C]$ for $AX^2 + BXY + CY^2$, and $f_I(p)$, $f_{II}(p)$ for the numbers of Type I and Type II solutions in the normalisation of [ET, §2].

2. Exact dictionaries

2.1. **The Elsholtz–Tao normal form.** For an odd prime p, p divides exactly one or exactly two of x, y, z in any solution of (1.1) (it cannot divide all three, since $1/x' + 1/y' + 1/z' = 4$ has no positive solution). Following [ET, §2] and [RT2, §2.0], put

$$V_I(p) = \{(a,b,c,d,e) \in \mathbb{N}^5 : \ 4abd = pe + 1, \ a + b = ce\},$$
$$V_{II}(p) = \{(a,b,c,d,e) \in \mathbb{N}^5 : \ 4abd = p + e, \ a + b = ce\}.$$

An element of $V_I(p)$ gives the solution $(abdp, acd, bcd)$, and an element of $V_{II}(p)$ gives $(abd, acdp, bcdp)$: indeed $(c + p(a+b))/(abcdp) = c(1+pe)/(abcdp) = 4/p$ and $(cp + a + b)/(abcdp) = c(p+e)/(abcdp) = 4/p$. Conversely every solution arises in this way [ET, Props. 2.2, 2.6]. Hence

$$\mathrm{ES}(p) \iff V_I(p) \cup V_{II}(p) \neq \emptyset. \tag{2.1}$$

On $V_I(p)$ one has $p \nmid cd$ [O130, Prop. 2.1], [O131, §5]. The counts $f_I(p)$, $f_{II}(p)$ count primitive tuples ($\gcd(a,b,c) = 1$) [ET, (2.10)]. A non-primitive tuple (ga', gb', gc', d) is the primitive tuple $(a', b', c', g^2 d)$, so existence questions are insensitive to primitivity.

2.2. **Type I: SL_2 matrices, Heegner wedges and the Hecke operator T_p.** The following is [O123, Lemmas 2.1–2.2, 2.4]; parts (i)–(ii) of Lemma 2.1 are rearrangements of [ET, §2].

Lemma 2.1 (PROVED)**.** *Let $a, c, d, f \geq 1$ with $p \nmid cd$. The following are equivalent:* (i) $p = 4acd - f$ *and* $f \mid 4a^2 d + 1$; (ii) $f \mid ap + c$ *and* $4ac \mid p + f$; (iii) $f \mid p^2 + 4c^2 d$ *and* $f \equiv -p \pmod{4cd}$. *In that case $e = (4a^2 d + 1)/f$, $b = (ap + c)/f$ and $g = (p^2 + 4c^2 d)/f = 4bcd - p$ are integers, $4abd - pe = 1$, $a + b = ce$, and*

$$\gamma = \begin{pmatrix} 4ad & p \\ e & b \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Proof. (i)⇒(ii): $c(4a^2 d + 1) = a \cdot 4acd + c = a(p + f) + c$, so $f \mid ap + c$. (ii)⇒(i): with $4acd = p + f$ the same identity gives $f \mid c(4a^2 d + 1)$, and $\gcd(f, c) \mid \gcd(4acd - p, c) \mid p$; if $p \mid f$ and $p \mid c$ then $p \mid cd$, which is excluded. (i)⇔(iii): modulo f we have $4acd \equiv p$, hence $4c^2 d(4a^2 d + 1) \equiv p^2 + 4c^2 d$, and $\gcd(f, 4c^2 d) \mid p$, which is 1 unless $p \mid cd$. For the identities, put $e = (4a^2 d + 1)/f$ and $b = ce - a$. Then $bf = c(4a^2 d + 1) - a(4acd - p) = ap + c$ (so $b \geq 1$), $4abd = e \cdot 4acd - 4a^2 d =$

$e(p+f)-4a^2d = pe+1$, and $(4acd-p)(4bcd-p) = 4c^2d(4abd-pe)+p^2 = p^2+4c^2d$, using $a+b=ce$; see [ET, (2.8)–(2.9)]. □

Write $Q_- = [f, -4ad, de]$ (discriminant $-4d$) and $Q' = [f, 4cd, dg]$ (discriminant $-4dp^2$).

Proposition 2.2 (Hecke form; PROVED)**.** *For every Type I tuple, $Q'(x,y) = Q_-(x+2bd\,y,\ p\,y)$, i.e. $Q' = Q_- \circ \left(\begin{smallmatrix} 1 & 2bd \\ 0 & p \end{smallmatrix}\right)$. Thus the discriminant-$(-4dp^2)$ form of a Type I solution is a T_p-Hecke translate of a level-d Heegner form of discriminant $-4d$, and the roots satisfy $z_{Q'} = p\, z_{Q_-} - 2bd$.*

Proof. The middle coefficient is $2f \cdot 2bd - 4adp = 4d(bf-ap) = 4cd$. The last is $4b^2d^2f - 8abd^2p + dep^2 = d\big(4bd(ap+c) - 8abdp + ep^2\big) = d\big(4bcd - p(4abd-pe)\big) = d(4bcd-p) = dg$. □

Proposition 2.3 (the wedge; PROVED, [O123, Prop. 3.1])**.** *Let $\Lambda_d = \{z = (2ad + i\sqrt{d})/f : a, f \geq 1,\ f \mid 4a^2d+1\}$, the roots of the forms $[f, -4ad, de]$. For $z = x+iy \in \Lambda_d$ and $c \geq 1$,*

$$\frac{2c\sqrt{d}\,x - \sqrt{d} - p\,y}{y} = 4acd - f - p \in \mathbb{Z}.$$

Hence (a,c,d,f) is a Type I tuple of p if and only if z lies in the wedge $R_c = \{x+iy : |2c\sqrt{d}\,x - \sqrt{d} - py| < y\}$ at the cusp $1/(2c)$. The edges of R_c are hypercycles of the geodesic $\{\mathrm{Re}\, z = 1/(2c)\}$, and R_c is a tube of hyperbolic width $\operatorname{arsinh}\frac{p+1}{2c\sqrt{d}} - \operatorname{arsinh}\frac{p-1}{2c\sqrt{d}} \approx 2/\sqrt{p^2+4c^2d}$.

Proof. For $z \in \Lambda_d$ we have $x/y = 2a\sqrt{d}$ and $\sqrt{d}/y = f$; substitute. An integer of absolute value < 1 is 0. The width is the difference of the distances of the two edges from the geodesic. □

The target is therefore at the *Planck scale* $\asymp 1/p$ of the relevant discriminant. The same holds in the T_p-translated picture, where the wedge sits at $-1/(2a)$ [O123, Prop. 3.2].

Lemma 2.4 (genus condition; PROVED; classical)**.** *If (a,c,d,f) is a Type I tuple of an odd n with $\gcd(n, 2cd) = 1$, then $\chi_{-4d}(-n) = 1$. For $n = p$ prime this says $\left(\frac{-d}{p}\right) = -1$: p is inert in $\mathbb{Q}(\sqrt{-d})$. For an odd square $n = m^2$ the condition fails for every d.*

Proof. Every prime $q \mid f$ divides $n^2 + 4c^2d$ and is prime to $2cd$, so $-4c^2d \equiv n^2$ is a non-zero square mod q, and $\chi_{-4d}(f) = 1$. As $f \equiv -n \pmod{4cd}$ and χ_{-4d} has period dividing $4d$, $\chi_{-4d}(-n) = \chi_{-4d}(f) = 1$. For $n = m^2$, $\chi_{-4d}(-m^2) = \chi_{-4d}(-1) = -1$. For $n = p$, $\chi_{-4d}(-p) = -\chi_{-4d}(p) = -\left(\frac{-d}{p}\right)$. □

This is the Type I half of [ET, Prop. 1.6], read at a fixed level. Any fixed-p main term at level d must therefore carry the factor $(1 + \chi_{-4d}(-p))/2$.

2.3. **Type II: determinant one at level dp, and shells.** The following is [O128, Lemma 1.1, Props. 2.1, 2.5, 7.1, Lemma 6.1].

Proposition 2.5 (PROVED)**.** *Let $c, d \geq 1$, $M = 4cd$ and $N_{c,d} = 4c^2dp + 1$.*

(a) *Type II tuples of p with these (c,d) correspond to factorisations $ff'-4c^2(dp)=1$ with $f\equiv f'\equiv -1 \pmod M$, via $f=4acd-1$, $f'=4bcd-1$. Equivalently, f runs over the divisors of $N_{c,d}$ with $f\equiv -1 \pmod M$ (the cofactor is then automatically $\equiv -1$).*

(b) *The map $(c,f)\mapsto Q=[f,4cdp,dpf']$ sends these solutions injectively to Heegner forms of discriminant $-4dp$ at level dp. For fixed (p,d) the corresponding points of $\mathbb{H}$ are pairwise at hyperbolic distance with $\cosh\geq 3/2$.*

(c) *In the determinant-p coordinates $Q=[f,4ad,de]$ (level one, discriminant $-4dp$), with $S_c=[0,2c,1]$ and the bilinear form $b(x,y)=q(x+y)-q(x)-q(y)$ of $q([A,B,C])=B^2-4AC$, one has $q(Q)=-4dp$, $q(S_c)=4c^2$ and $b(Q,S_c)=4$. Thus the Type II solutions are embeddings of the binary lattice $4\cdot[-dp,1,c^2]$ (discriminant $4c^2dp+1$ before scaling) into the trace-zero lattice of signature $(2,1)$. Geometrically, a Type II point is a Heegner point of discriminant $-4dp$ at hyperbolic distance exactly $\operatorname{arsinh}\big(1/(2c\sqrt{dp})\big)$ from the cuspidal geodesic $(\infty,-1/(2c))$: it lies on a* shell.

(d) *(genus lemma) $(p/f)=(p/f')=-1$ (Jacobi symbols).*

Proof of (a), (c), (d). (a) $(4acd-1)(4bcd-1)=4c^2d(4abd-e)+1=4c^2dp+1$ using $a+b=ce$ and $4abd=p+e$. Conversely, given $f\mid N_{c,d}$ with $f\equiv -1 \pmod M$, the cofactor f' satisfies $ff'\equiv 1$, hence $f'\equiv -1 \pmod M$; put $a=(f+1)/M$, $b=(f'+1)/M$; expanding $ff'=N_{c,d}$ gives $4abcd=a+b+cp$, so $c\mid a+b$ and $e=(a+b)/c=4abd-p$. (c) With $e=(p+4a^2d)/f$ (an integer, [ET, (2.19)]), $q(Q)=16a^2d^2-4def=-4dp$ and $b(Q,S_c)=2\cdot 4ad\cdot 2c-4f=4(4acd-f)=4$. The distance formula is [O128, Prop. 2.5]. (d) $ef=p+4a^2d$ gives $(2ad)^2\equiv -dp \pmod f$ with $\gcd(dp,f)=1$, so $(-dp/f)=1$. Since $f\equiv -1 \pmod{4d}$ and $\chi_{-4d}(-1)=-1$, $(-d/f)=\chi_{-4d}(f)=-1$, hence $(p/f)=-1$. The same argument applies to f'. □

Part (b) is [O128, Prop. 2.1, Lemma 2.2]; the separation is the lemma $\cosh\operatorname{dist}-1=\operatorname{disc}(Q-Q')/(8dp)$ of [TH, Lemma 2.2] applied at level dp. Part (d) recovers the Type II half of [ET, Prop. 1.6]: at an odd square $n=s^2$ one has $(s^2/f)=1$. As for Type I, it is a level-wise reading of a classical fact; see Proposition 2.6.

2.4. **The Brauer class is a genus character (folklore).** Bright and Loughran [BL] study the surface $U_n: 4u_1u_2u_3=n(u_1u_2+u_1u_3+u_2u_3)$ and its integral model. Over $\mathbb{Q}$ the group $\operatorname{Br}\tilde{U}_n/\operatorname{Br}\mathbb{Q}$ is $\mathbb{Z}/2$, generated by the transcendental quaternion algebra $\alpha=(-u_1/u_3,-u_2/u_3)$ [BL, Cor. 2.7]. They show that there is no Brauer–Manin obstruction to positive integral points for $n\geq 2$ [BL, Thms. 1.1, 1.8], that α obstructs strong approximation [BL, Thms. 1.8, 1.9], and that it recovers the odd-square obstruction [BL, Thm. 1.2, Cor. 1.4].

Proposition 2.6 (PROVED; folklore, Yamamoto [Ya] via [BL, App. A])**.** (a) *On the Type I locus at level d, the condition imposed by α is $\left(\frac{-ab}{p}\right)=\left(\frac{-d}{p}\right)=-1$, i.e. Lemma 2.4. For $\left(\frac{-d}{p}\right)=+1$ the positive adelic Brauer set of the level-d chart is empty.* (b) *On the Type II locus, α imposes $(f/p)=-(-1/p)$, which is equivalent to Proposition 2.5(d) and holds automatically. So the Brauer–Manin obstruction never removes a Type II level.*

Sketch. (a) $4abd=pe+1$ gives $(abd/p)=1$, so $(-ab/p)=(-d/p)$. Lemma 2.4 gives $(-d/p)=-1$. For the local statement, $\operatorname{inv}_p\alpha=(-ab/p)$ on p-adic points of the chart,

$\mathrm{inv}_\infty \alpha = -1$ on the positive part, and $\mathrm{inv}_\ell \alpha = +1$ for $\ell \neq p$ [BL, Lemmas 3.1, 3.5, 3.8]. (b) $e \equiv 4abd \pmod p$ gives $(-ab/p) = (-de/p)$, and $ef \equiv 4a^2 d \pmod p$ gives $(e/p) = (d/p)(f/p)$. Since $f \equiv 3 \pmod 4$, reciprocity gives $(f/p) = (p/f)(-1/p)$. Details: [O126, Props. 5.1, 5.2]. □

Nothing here is new. Part (a) is the classical reason why the Mordell/Elsholtz–Tao families $n \equiv -f \pmod{4ad}$, $f \mid 4a^2 d + 1$, cover only non-residue classes. Part (b) is Yamamoto's lemma $(p/q) = -1$ for $q = 4abd - 1$ in Mordell's notation, as derived from α in [BL, App. A]. (Yamamoto's paper was not accessed directly.)

2.5. **Quaternion orders.** For $d, m \geq 1$ let $\Lambda_{d,m} = \mathbb{Z}\langle u, v\rangle$ with $u^2 = -d$, $v^2 = -m$, $uv + vu = 1$. It is an order $\mathbb{Z} + \mathbb{Z}u + \mathbb{Z}v + \mathbb{Z}uv$ in the definite algebra $B_{d,m} \cong (-d, -d(4dm-1))_{\mathbb{Q}}$ (replace v by $2dv + u$, which anticommutes with u), of reduced discriminant $4dm - 1$: the Gram matrix of the reduced trace on $(1, u, v, uv)$ has determinant $-(4dm-1)^2$ [O130, Lemma 1.2].

Proposition 2.7 (PROVED; [O130, Prop. 1.3])**.** *Let $(a, b, c, d, e) \in V_I(p)$ and $m = ab$. Then $B_{d,m} \cong B_{p,\infty}$, the quaternion algebra ramified exactly at p and ∞, and $\Lambda_{d,m}$ is an order of reduced discriminant $4abd - 1 = pe$ in it. Consequently there is a supersingular elliptic curve over $\overline{\mathbb{F}}_p$ with endomorphisms u, v satisfying $u^2 = -d$, $v^2 = -ab$, $uv + vu = 1$.*

Sketch. Compute the Hilbert symbols of $(-d, -dpe)$. At $\ell \nmid 2dpe$ they are trivial. At p: $p \nmid e$ (if $p \mid e$ then $-4a^2 d \equiv 1 \pmod p$, contradicting $(-d/p) = -1$), so the symbol is $(-d/p) = -1$ by Lemma 2.4. At odd $\ell \mid e$: $-d \equiv (2a)^{-2}$ is a square mod ℓ ($e \mid 4a^2 d + 1$), so the symbol is 1. At odd $\ell \mid d$: $pe \equiv -1 \pmod \ell$ gives 1. At ∞ it is -1, and the product formula gives 2. The last sentence is Deuring's correspondence. □

This was checked on all Type I tuples with $p < 3000$ (EVIDENCE, [O130]). In the language of Gross–Zagier [GZ], the identity $(16abd - 4)/4 = pe$ has the shape $(D_1 D_2 - x^2)/4$ with $D_1 = -4d$, $D_2 = -4ab$, $x = 2$, so the CM points of these discriminants have a common supersingular reduction. This is a *formal analogy only*: the theorem of [GZ] assumes coprime fundamental discriminants, and $-4d, -4ab$ are never coprime. No Gross–Zagier formula is used.

Lemma 2.8 (PROVED; [O130, Lemma 3.1])**.** *In $B = (-d, -p)_{\mathbb{Q}}$ with $i^2 = -d$, $j^2 = -p$, $k = ij = -ji$, Type II at level d holds if and only if, for some $a, b, c, e \geq 1$,*

$$(2a\,i + j)(2b\,i - j) = -e(1 + 2c\,k).$$

Proof. The left side is $-4abd - 2(a+b)k + p = -(4abd - p) - 2(a+b)k$; compare coefficients with $4abd - p = e$, $a + b = ce$. □

Taking reduced norms gives $\mathrm{N}(2ai + j) = 4a^2 d + p = ef$ and $\mathrm{N}(1 + 2ck) = 1 + 4c^2 dp = ff'$, the integers of the ternary form below and of the shells of Proposition 2.5. The novelty of the quaternion dictionary is *unverified*: the survey [LA] found no prior occurrence, but we do not claim it is new.

2.6. **The CM torus.** Let $p \nmid 2cd$, $K = \mathbb{Q}(\sqrt{-d})$, $\alpha = p + 2c\sqrt{-d}$, so $\mathrm{N}\,\alpha = p^2 + 4c^2 d$.

Proposition 2.9 (PROVED; [O131, Lemma 5.1, Thm. 5.2])**.** (i) $\mathfrak{a} \mapsto \mathrm{N}\,\mathfrak{a}$ *is a bijection from the ideal divisors of (α) in $\mathbb{Z}[\sqrt{-d}]$ to the positive divisors of $\mathrm{N}\,\alpha$.* (ii) *The*

number $T_{c,d}(p)$ of ordered Type I tuples with these (c,d) equals $\#\{\mathfrak{a} \mid (\alpha) : \mathrm{N}\,\mathfrak{a} \equiv -p \pmod{4cd}\}$, and

$$T_{c,d}(p) = \frac{1}{\varphi(4cd)} \sum_{\chi \bmod 4cd} \bar{\chi}(-p)\, \sigma_\chi(p^2 + 4c^2 d), \qquad \sigma_\chi(N) = \sum_{f \mid N} \chi(f).$$

(iii) *Only Hecke characters of the form $\chi \circ \mathrm{N}$ occur. In particular every class-group character of K that is not a genus character has coefficient 0, and pairing $\chi \leftrightarrow \chi\chi_{-4d}$ produces the factor $1 + \chi_{-4d}(-p)$ of Lemma 2.4.*

Sketch. α is primitive since $\gcd(p, 2c) = 1$. Every prime $\ell \mid \mathrm{N}\,\alpha$ is odd and prime to d, so it is unramified and not inert, and it splits. Exactly one of the two primes above ℓ divides α, to the full power. This gives (i). Then (ii) is Lemma 2.1(iii) plus orthogonality on $(\mathbb{Z}/4cd)^\times$, and (iii) is genus theory. □

Thus the dihedral (Frobenius) information carried by the class of (f, α) in the ring class field is invisible in the count.

2.7. **The isotropic ternary form Q_d.** Let $Q_d(e, f, a) = ef - 4da^2$.

Proposition 2.10 (PROVED; [O124, Prop. 6.2], [RT1, §6])**.** (i) *Type II at level d with parameter a holds if and only if $Q_d(e, f, a) = p$ with $4ad \mid f + 1$ and $(f+1)e/(4ad) > a$, i.e. $p + 4a^2 d$ has a divisor $f \equiv -1 \pmod{4ad}$ with large enough cofactor.* (ii) *Q_d represents every integer primitively over $\mathbb{Z}$ ($(e, f, a) = (n, 1, 0)$), so local conditions, genus and spinor genus impose nothing; all the arithmetic sits in the side condition $4ad \mid f + 1$.* (iii) *The side condition is not a finite union of lattice cosets: its slice at $a = a_0$ has f-period $4a_0 d$, which is unbounded.* (iv) *On such points the cofactor $f = 4acd - 1$ is a Type II* multiplier witness modulus*; in particular $f \geq W(p)$, the least such modulus.*

Among all the parametrisations of [ET, §2], Q_d is the only rank-3 quadratic piece (CERTIFIED by a systematic search, [O124, Comp. 6.1], reproduced independently in its review).

3. The divisor-class reduction

This section is [RT2, §2]. A *slice* is a choice θ of two of the coordinates $\{a, b, c, d, e\}$ together with prescribed values for them; $V_\theta(p)$ denotes the elements of $V_I(p)$ (or $V_{II}(p)$) with these values. By the symmetry $a \leftrightarrow b$ the coordinate pairs are $\{a,b\}, \{a,c\}, \{a,d\}, \{a,e\}, \{c,d\}, \{c,e\}, \{d,e\}$.

Lemma 3.1 (slice table; PROVED, elementary)**.** *For each type and each slice θ exactly one of the following holds.*

- (C) Congruence slice: *$V_\theta(p) \neq \emptyset$ if and only if p lies in an explicit finite union of residue classes modulo a modulus depending only on θ.*
- (D) Divisor slice: *there are an integer $N_\theta(p)$ (a polynomial in p of degree ≤ 2, possibly divided by a constant depending on θ), a modulus q_θ and a residue set $R_\theta(p) \subset \mathbb{Z}/q_\theta$ such that $V_\theta(p)$ is in bijection with $\{f \geq 1 : f \mid N_\theta(p),\ f \bmod q_\theta \in R_\theta(p)\}$. The set $R_\theta(p)$ depends only on θ, on $p \bmod q_\theta$ and, in the $\{d, e\}$ rows, on $N_\theta(p) \bmod q_\theta$.*

The rows are as follows (Type I rows containing c or d are for $p \nmid cd$; otherwise the fibre is empty).

type	θ	*kind*	$N_\theta(p)$	*divisor* f	*class condition*
I	$\{c,d\}$	*D*	p^2+4c^2d	$4acd-p$	$f\equiv -p\ (4cd)$
I	$\{a,c\}$	*D*	$ap+c$	$4acd-p$	$f\equiv -p\ (4ac)$
I	$\{a,e\}$	*D*	$(pe+1)/4a$	b	$f\equiv -a\ (e)$
I	$\{d,e\}$	*D*	$m=(pe+1)/4d$	a	$f^2\equiv -m\ (e)$, $(f,e)=1$
I	$\{a,d\}$	*C*	$4a^2d+1$	$4acd-p$	$p\equiv -f\ (4ad)$
I	$\{a,b\},\{c,e\}$	*C*	—	—	p *mod* $4ab$, *resp.* $4a(ce-a)$
II	$\{c,d\}$	*D*	$4c^2dp+1$	$4acd-1$	$f\equiv -1\ (4cd)$
II	$\{a,c\}$	*D*	$cp+a$	$4acd-1$	$f\equiv -1\ (4ac)$
II	$\{a,d\}$	*D*	$p+4a^2d$	$4acd-1$	$f\equiv -1\ (4ad)$
II	$\{a,e\}$	*D*	$(p+e)/4a$	b	$f\equiv -a\ (e)$
II	$\{d,e\}$	*D*	$m=(p+e)/4d$	a	$f^2\equiv -m\ (e)$, $(f,e)=1$
II	$\{a,b\},\{c,e\}$	*C*	—	—	p *mod* $4ab$, *resp.* $4a(ce-a)$

In the $\{a,e\}$ *and* $\{d,e\}$ *rows,* $N_\theta(p)$ *is required to be an integer (otherwise the fibre is empty).*

Proof. Each D row is an identity plus a converse. $I\{c,d\}$ is Lemma 2.1(iii). For the converse, the cofactor $f'=N/f$ satisfies $f'\equiv p^2(-p)^{-1}\equiv -p \pmod{4cd}$; put $a=(f+p)/4cd$, $b=(f'+p)/4cd$; expanding $ff'=N$ gives $p(a+b)=c(4abd-1)$, so $c\mid a+b$ and $e=(a+b)/c=(4abd-1)/p$. $II\{c,d\}$ is Proposition 2.5(a). $I\{a,c\}$ is Lemma 2.1(ii). $II\{a,c\}$, $II\{a,d\}$: with $f=4acd-1$ one has $ef=4a^2d+p$, and $c(4a^2d+p)=a(f+1)+cp$, so $f\mid cp+a$ (as $\gcd(f,c)=1$). Conversely put $d=(f+1)/4ac$, resp. $c=(f+1)/4ad$, then $e=(p+4a^2d)/f$ and $b=ce-a$. $\{a,e\}$, $\{d,e\}$: with $M=pe+1$ (Type I) or $M=p+e$ (Type II) the defining equations read $bd=M/4a$, resp. $ab=M/4d$, together with $e\mid a+b$. For $\gcd(a,e)=1$, $e\mid a+m/a$ is equivalent to $e\mid a^2+m$, and every element has $\gcd(a,e)=1$: for Type I since $a\mid pe+1$; for Type II since $\gcd(a,e)\mid p$ and $p\nmid e$ (if $p\mid e$ then $4abd=p+e\le 2e\le 2(a+b)$ forces $a=b=d=1$ and then $4=p+e\ge 2p$, impossible). *C rows*: for I$\{a,d\}$ the fibre is $\{f\mid 4a^2d+1 : f\equiv -p\ (4ad)\}$, a condition on p mod $4ad$; for $\{a,b\}$, e runs over the divisors of $a+b$ and the condition is $4ab\mid pe+1$ (resp. $p+e$); for $\{c,e\}$, a runs over $[1,ce-1]$ with condition $4a(ce-a)\mid pe+1$ (resp. $p+e$). □

All bijections were verified mechanically against a brute-force enumeration of $V_I(p)\cup V_{II}(p)$ for every prime $5\le p<200$ (CERTIFIED, [RT2, Lemma 2.1]; script `scripts/rt2_slices.py`).

Theorem 3.2 (divisor-class reduction)**.** *Consider the encodings of §2: E1 the Type I* SL_2*/Heegner wedge (Proposition 2.3,* $\theta=\{c,d\}$*; its* T_p*-translated form, per tube,* $\theta=\{a,d\}$*); E2 the Type II shells (Proposition 2.5,* $\theta=\{c,d\}$*); E3 the quaternion order of Proposition 2.7 (*$\theta=\{d,e\}$*); E4 the* $(-d,-p)$ *factorisation of Lemma 2.8 (*$\theta=\{d,e\}$*; the norms of* $2ai+j$ *give* $\theta=\{a,d\}$*); E5 the CM torus (Proposition 2.9,* $\theta=\{c,d\}$*); E6 the ternary form* Q_d *(Proposition 2.10,* $\theta=\{a,d\}$*).*

(i) (PROVED) *The E-solutions with parameter* θ *are in bijection with* $V_\theta(p)$*. Hence they correspond to the divisors of the single integer* $N_\theta(p)$ *lying in* $R_\theta(p)$*, or (C rows) to a congruence condition on* p*.*

(ii) (PROVED, RELATIVE TO THE CITED INTERNAL STATEMENTS) *Every representative-independent invariant of these objects is congruence or genus data. Specifically: for E5 only genus characters have non-zero coefficient (Proposition 2.9). For E3 the collision datum is a function of* θ*,*

of $p \bmod 4d$, of $(-d/p)$ and of $(-d/\ell)$ for $\ell \mid e$, and it holds automatically on every admissible θ [O130, Prop. 2.2]*. For E1, E2 and E6 the property of being an ES point belongs to the representative, not to the CM point or Γ-orbit: each class has infinitely many representatives of the required shape and only finitely many good ones (*[O126, Prop. 4.1]*;* [O128, Prop. 8.2]*; the stabiliser of S_c in $\mathrm{SL}_2(\mathbb{Z})$ is $\pm I$). For E4 the ES elements lie on the line $\{2xi \pm j\}$, on which the reduced norm is the one-variable polynomial $4dx^2 + p$* [O130, Prop. 3.2].

(iii) (CONDITIONAL ON HYPOTHESIS H) *Let Θ_0 be a finite set of slice parameters, of any types and rows, fixed independently of p. Any argument that certifies $\mathrm{ES}(p)$ by exhibiting an E-solution with parameter in Θ_0 fails for infinitely many primes $p \equiv 1 \pmod{24}$. This holds whatever auxiliary automorphic, spectral or geometric information the argument uses.*

Proof. (i) Each source file proves the exact lift from its object to $V_\theta(p)$ (Propositions 2.3, 2.5, 2.7, 2.9, 2.10 and Lemma 2.8); then apply Lemma 3.1. (ii) is a collection of the cited statements. (iii) The argument can only succeed at primes with $\bigcup_{\theta\in\Theta_0} V_\theta(p) \neq \emptyset$. By Lemma 3.1 this predicate is decided, together with a witness, by the bounded and correct procedure "for $\theta \in \Theta_0$: test $p \bmod q_\theta$; factor $N_\theta(p)$ (with floor division for the $\{\cdot, e\}$ rows); test each divisor against $R_\theta(p)$". By [PS, Thm. C], under Hypothesis H every such procedure fails for infinitely many $p \equiv 1 \pmod{24}$. □

Remark 3.3 (scope)**.** Part (iii) says that whatever an automorphic input "knows", if it certifies ES through a fixed finite set of fibres it is no stronger than factoring finitely many fixed polynomials in p, and under H that fails. It does *not* say that automorphic methods cannot aggregate over a parameter set $\Theta(p)$ that grows with p; Kuznetsov's formula and Heegner equidistribution aggregate over levels and discriminants in exactly this way. Aggregation over θ is legitimate, since θ is a coordinate of the solution and not fixed by the problem. By Lemma 3.1 it amounts to controlling, at one prime p, the divisor events of a growing family $\{N_\theta(p)\}_{\theta\in\Theta(p)}$ jointly. The quantitative obstacles to doing so are in §§4–5, and the open routes in §6. Unconditional quantitative versions of (iii) exist for specific growing boxes. For the Type II box $\{acd \le T/4\}$ one uses $f = 4acd - 1 \ge W(p)$ together with the Ω-result $W(p) > \exp\big(c(\log p)^{1/4}(\log\log p)^{-1/4}\big)$ for infinitely many p [SE]. For Type I heights there is $ck_{\min}(p) \gg \log p \cdot \log\log\log p$ infinitely often (internal file `POINTWISE_OMEGA.md`, Thm. 8.5, modulo Lau–Wu). Both are cited as summarised in [RT2, §2.2].

4. Negative results with exact scope

Each result in this section concerns a precisely defined mechanism. None of them says that ES is false or unprovable.

4.1. **Ternary and higher-rank embeddings.**

Fix $k \ge 3$, a non-degenerate integral quadratic form Q in k variables, and integers $\kappa \ge 1$, η, all independent of p.

Definition 4.1 ([O124, Def. 1.1])**.** An *embedding of complexity* (J, D) is a family of pieces $(S_j, \Phi_j)_{j\le J}$, where $S_j \subseteq \mathbb{Z}^k$ is an *arbitrary* subset and $\Phi_j = (X_j, Y_j, Z_j)$ is a triple of rational functions of degree $\le D$. It is *valid* if for every prime p, every

j and every $v \in S_j$ with $Q(v) = \kappa p + \eta$, $\Phi_j(v) \in \mathbb{N}^3$ solves (1.1) for $n = p$. Put $R(p) = \bigcup_j \{v \in S_j : Q(v) = \kappa p + \eta\}$. A piece is *degenerate* if $X_j, Y_j, Z_j \in \mathbb{Q}(Q)$.

This covers every statement of the form "$\kappa p + \eta$ is represented by Q with side conditions, hence ES(p) by an explicit piecewise rational recipe" of fixed complexity. It does not cover recipes of unbounded algebraic complexity, or families with a parameter that grows with p.

Theorem 4.2 (PROVED, relative to Pila [Pi], Mordell–Schinzel [Mo, Sch] and [ET, Thm. 1.1]; [O124, Thm. 4.1, Cor. 4.2])**.** *Let Q be positive definite. There is a finite union U of primitive classes modulo some q^*, containing no square class, such that:*

(a) *for every large prime $p \notin U$, $\#R(p) \ll f(p)\, p^{(k-2)/2-1/4+\varepsilon}$, where $f(p)$ is the total number of solutions;*
(b) *for $0 < \varepsilon_0 < 1/4$ the primes $p \le N$, $p \notin U$, with $\#R(p) \ge c p^{(k-2)/2-\varepsilon_0}$ number $\ll N^{3/4+\varepsilon_0+\varepsilon}$.*

Consequently, suppose a representation theorem (circle method for $k \ge 5$, Kloosterman/δ-method for $k = 4$, Duke [Du], Duke–Schulze-Pillot [DSP] and Blomer-type results for $k = 3$) gives $\#R(p) \ge c p^{(k-2)/2-\varepsilon_0}$ on a union $\mathcal{P}_R$ of primitive classes, outside $o(\pi(N))$ exceptions. Then $\mathcal{P}_R \subseteq U$: the representation input proves ES only for primes that a finite polynomial (Mordell-type) covering already handles.

Sketch. A degenerate piece produces an ES solution depending only on p. By a rationality and integrality argument it is a polynomial identity on finitely many classes, which by Mordell–Schinzel are not square classes; this defines U. A non-degenerate piece is non-constant on the quadric $Q = m$, $m = \kappa p + \eta$, outside finitely many p. Group the points of $R_j(p)$ by the value of a non-constant coordinate X_j; there are at most $f(p)$ values. Each group lies on a hypersurface section of the bounded quadric $Q = m$. For $k = 3$ such a section is a curve whose real components have degree ≥ 2 (no real line lies on a bounded quadric). By Pila's bound it has $\ll m^{1/4+\varepsilon}$ integer points in a box of side $\asymp m^{1/2}$. For $k \ge 4$ one slices by the last $k-3$ coordinates. This gives (a). Then (b) follows from (a) and the first-moment bound $\sum_{p \le N} f(p) \ll N \log^2 N \log\log N$ [ET, Thm. 1.1], and the corollary from Dirichlet's theorem. □

The same holds for line-free indefinite ternary forms [O124, §5]. The theorem does *not* say that an embedding proves ES only on U. A non-degenerate piece supported on a *thin* set can cover a positive proportion of primes. An example from [O124, §4]: $Q = \sum_{i \le 5} v_i^2$, with $Q(v) = p + 1$, the piece $v_1 = 1$, $4 \mid Q(v)$, and the output $((p+1)/4 + v_1 - 1,\ p(p+1)/2,\ p(p+1)/2)$. What it excludes is that a representation *theorem*, which produces only sets of the generic size, does any work outside U. The only rank-3 quadratic piece of ES is Q_d (Proposition 2.10). It represents everything and is isotropic, and its side condition is not mild. EVIDENCE: every Mordell-hard prime $\le 10^8$ is solved through Q_d with $d \le 8$, $a \le 22$ [O124, §8].

4.2. **The scale constraint.**

Definition 4.3 ([O127, Def. 2.2])**.** An *ES-detector* assigns to each large prime p a finite target $\mathcal{T}_p$ and a map $\Phi_p : \mathcal{T}_p \to \mathrm{Sol}(p)$ whose fibres have size $\le \mu(p)$. A hypothesis H *drives* it if H implies $\#\mathcal{T}_p \ge M(p) - R(p)$ with explicit $M(p) > R(p)$.

Proposition 4.4 (PROVED, relative to [ET, Thm. 1.1, Prop. 1.7]; [O127, Prop. 2.3], [RT1, Prop. 2.4, §5.5])**.** *If* H *drives a detector with* $R(p) \le M(p)/2$*, then* $M(p) \le 2\mu(p)f(p)$*. Here*

(a) $f(p) \le p^{3/5+O(1/\log\log p)}$ *for every* p [ET, Prop. 1.7]*;*
(b) *for each* $\delta > 0$*,* $f(p) \le p^{\delta}$ *outside a set of primes with counting function* $O_\delta(N^{1-\delta}\log^3 N)$*;*
(c) *for any* $\xi(p) \to \infty$*,* $f(p) \le \xi(p)\log^3 p \log\log p$ *outside a set of primes of relative density* $o(1)$*.*

In particular, if $\mu(p) \le p^{o(1)}$*, then* $M(p) \le p^{o(1)}$ *for almost all* p*, and no detector with* $M(p) \ge p^{\eta}$ *for all large* p *can be driven.*

Proof. $M/2 \le \#\mathcal{T}_p \le \mu(p)f(p)$. Part (a) is [ET, Prop. 1.7]. Parts (b) and (c) follow from Markov's inequality applied dyadically to $\sum_{p\le N} f(p) \ll N\log^2 N \log\log N$ [ET, Thm. 1.1]. □

Every natural parametrisation of solutions by lattice points, Heegner points or matrices is $O(1)$-to-one, or $p^{o(1)}$-to-one by the divisor bound. So any equidistribution statement whose main term for the target is a power of p is excluded as a driver, unless it detects only a coarse invariant of solutions (fibres $\mu \ge p^{\eta-o(1)}$). In that case the lifting step from the invariant to an integer solution must be supplied separately. This applies in particular to small-scale equidistribution of Heegner or CM points in families with $|D| \ge p^{\kappa}$ through a $p^{o(1)}$-to-one encoding [O127, Cor. 4.1]. With the Type I bound of [TH] in place of [ET, Thm. 1.1], Markov's inequality gives $f_I(p) \le \psi\mathcal{L}^3$ for all but a proportion $O(1/\psi)$ of primes [O123, Thm. 5.1]. A Planck-scale certificate must therefore have *polylogarithmic* accuracy.

Proposition 4.5 (local data of moderate modulus; PROVED; [O127, Prop. 7.3], [RT1, Lemma 5.1])**.** *Let* $S'(n)$ *count the solutions of the* $a = 1$ *Type I slice* $(tn+1 = mm'',\ m \equiv m'' \equiv -1 \pmod t,\ 4 \mid m)$ *with* $\gcd(n,c) = 1$*. Let* $\mathsf{H}(n)$ *be any predicate depending only on the dyadic range* $n \in (X, 2X]$ *and on* $n \bmod Q(X)$*, where* $Q(X) \le X^{1/2-\varepsilon}$*. If* H *holds at one large prime* $p \in (X, 2X]$ *that is a square modulo* $Q(X)$*, then it holds at an odd square in* $(X, 2X]$*, where* $S' = 0$*. Hence "*$\mathsf{H}(n) \Rightarrow S'(n) \ge 1$*" is false for any such* H*.*

Proof. The odd $m \equiv m_0 \pmod{2Q}$ with $m_0^2 \equiv p \pmod Q$ give $\asymp X^{1/2}/Q$ squares in the class of p. For odd $n = m^2$, a slice solution with $\gcd(n,c) = 1$ is a Type I solution, which cannot exist by [ET, Prop. 1.6]. □

This does not apply to "$\mathsf{H}(n) \Rightarrow \mathrm{ES}(n)$": odd squares satisfy ES, through solutions that are of neither type.

4.3. **Thickening a target is averaging over** n**.**

Proposition 4.6 (PROVED; [O123, Prop. 5.4])**.** *In the setting of Proposition 2.3, for* $z \in \Lambda_d$ *the integer* $k(z) = (2c\sqrt{d}\,x - \sqrt{d} - py)/y = 4acd - f - p$ *equals* $n - p$*, where* $n = 4acd - f$ *is the integer for which* (a, c, d, f) *satisfies Lemma 2.1(i) with* p *replaced by* n*. Hence the thickened target* $R_c^{(K)} = \{|2c\sqrt{d}\,x - \sqrt{d} - py| \le Ky\}$*, of hyperbolic width* $\approx (2K+1)/p$*, counts exactly*

$$\sum_{|n-p|\le K} \#\{(a,f):\ f \mid 4a^2d+1,\ 4acd - f = n,\ f \le 2p\}.$$

Proof. The computation of Proposition 2.3, with $k(z) \in \mathbb{Z}$ allowed to range over $[-K, K]$. □

The counted points are Elsholtz–Tao points of the integers n near p. For composite n the coprimality $\gcd(c, n) = 1$ is not imposed, so the count is at least the Type I count of n with these (c, d), not equal to it. Assessment. A spectral estimate at Sobolev scale V (target width $\asymp 1/V$) therefore controls only averages of Type I counts over intervals $|n - p| \lesssim p/V$. With V a small power of the discriminant (§5) these are intervals of length $p^{1-\kappa}$. In that range unconditional exceptional-set bounds are already available, and nothing pointwise follows. The pointwise problem is exactly $K = 0$.

4.4. **Local, genus and spinor solubility hold at odd squares.** In the lattice language of Proposition 2.5(c), Type II at (c, d) for an integer n asks for $Q = [A, B, C] \in L = \mathbb{Z}^3$ with $q(Q) = -4dn$, $b(Q, S_c) = 4$, $B \equiv 0 \pmod{4d}$, $d \mid C$ and $A > 0$.

Proposition 4.7 (proved; [O128, Prop. 8.3])**.** *For every $n \geq 1$, including odd squares, and every (c, d), the vector $Q_0 = [-1, 0, -dn]$ satisfies every condition above except $A > 0$: $q(Q_0) = -4dn$, $b(Q_0, S_c) = 4$, $B \equiv 0 \pmod{4d}$, $d \mid C$. Hence the Type II embedding problem is everywhere locally soluble, and even globally soluble without the sign condition, for every n. For odd squares $n = s^2$ it has no solution with $A > 0$ (Proposition 2.5(d)).*

Proof. $q(Q_0) = 0 - 4(-1)(-dn) = -4dn$ and, since $b(Q, S_c) = 4(cB - A)$, $b(Q_0, S_c) = 4$. Solubility at every prime ℓ is witnessed by Q_0. At ∞ take $B > 0$ large, $A = cB - 1 > 0$ and $C = (B^2 + 4dn)/(4A) > 0$. The last sentence is [ET, Prop. 1.6]. □

So no statement "everywhere locally soluble ⇒ soluble" holds for this integral-point problem. Q_0 is the trivial factorisation $N = (-1)(-N)$ of $N = 4c^2dn + 1$, i.e. $a = 0$. The content of Type II is that N has a positive divisor in the class $-1 \pmod{4cd}$, while it always has the positive divisor 1 in the class $+1$. The two differ only by the sign at ∞. The only restriction forced locally is the genus character $\chi_{-4d} \cdot (n/\cdot)$, a $\mathbb{Z}/2$ quotient that is non-trivial exactly at squares [O128, Prop. 9.2]. There is no natural spinor genus for this pair problem, and the phrase "spinor obstruction" does not apply. Once S_c is fixed, the stabiliser is a *split* torus, and the problem is representation by $x^2 - y^2$ in a coset, i.e. a divisor-in-class question [O128, §8]. This is the reason (an Assessment) why the Gauss/three-squares mechanism (class number one of a ternary genus, with an anisotropic stabiliser) does not transfer.

4.5. **Geometry: fibrations, strong approximation, Brauer–Manin.** Let U_n and α be as in Proposition 2.6, and let L_{ij} be the lines $u_i = u_j = 0$ and M_k the lines $u_k = n/4$, $u_i = -u_j$ (rational only if $4 \mid n$).

Theorem 4.8 (proved, relative to [BL, Lemma 3.10]; [O126, Thm. 3.1])**.** *Let $C \subset U_n$ be an irreducible curve over $\mathbb{Q}$ containing infinitely many integral points. Then C is one of the lines L_{ij}, or one of the M_k (only if $4 \mid n$; these carry no positive points). In particular, for every dominant map $U_n \to B$ to a curve, all fibres except finitely many lines have finitely many integral points, and no fibre has infinitely many positive integral points.*

Proof. Points with a zero coordinate lie on the L_{ij}. By [BL, Lemma 3.10], integral points with $u_1u_2u_3 \neq 0$ and $u_i \neq -u_j$ are finite in number. Points with $u_i = -u_j \neq 0$ satisfy $u_k = n/4$ and have a negative coordinate. □

The reason is the real place, not the splitness of the fibres: the open torus $V_n \subset \tilde{U}_n$ of [BL, Lemma 2.1] is split over $\mathbb{Q}$ [O126, Prop. 3.2]. No fibration argument can produce infinitely many positive points in a fibre.

Proposition 4.9 (PROVED; [O126, §1], [BL])**.** *For every $n \geq 2$, the statement "the Brauer–Manin obstruction is the only obstruction to positive integral points on U_n" is equivalent to* ES(n)*. For $n = 1$ the Brauer–Manin obstruction does obstruct.*

Sketch. For $n \geq 2$ the positive Brauer set is non-empty by [BL, Thm. 1.8], so "BM is the only obstruction" says exactly that a positive integral point exists. For $n = 1$ there is no positive solution, and every point of the positive adelic set has $\mathrm{inv}_\infty \alpha = -1$ and $\mathrm{inv}_\ell \alpha = +1$ for all finite ℓ [BL, Lemmas 3.1, 3.5, 3.8], so the Brauer set is empty [O126, §1]. (The proof of [BL, Thm. 1.8] needs a prime divisor of n; their Thm. 1.1 correctly assumes $n \geq 2$.) □

So a "BM is the only obstruction" conjecture for this family is ES restated.

On the homogeneous spaces $X_d : \ ef - 4da^2 = 1$ ($\cong \mathrm{Spin}(2,1)/T$, where strong approximation holds), the Type I locus is $\{ef - 4da^2 = 1,\ f + p = 4cda,\ c \neq 0\}$. The ES condition $4da \mid f + p$ is a congruence modulo a *coordinate of the point*, not modulo a fixed modulus. So it is not an adelic open condition away from ∞, and strong approximation does not reach it. For each (p, d) the locus has finitely many integral points, while $X_d(\mathbb{Z})$ is infinite [O126, Prop. 4.1] (PROVED). That descent and strong approximation are irrelevant beyond this is an ASSESSMENT [O126, Cor. 4.2]. The same shape holds for Type II on Q_d and on the level-dp shells.

4.6. **Single targets and fixed shells are empty infinitely often.**

Proposition 4.10 (CONDITIONAL on Hypothesis H; [O123, Prop. 5.3])**.** *There are infinitely many primes $p \equiv 5 \pmod{12}$ for which the genus-allowed Type I target $(c, d) = (1, 3)$ is empty, although its expected size is $\approx 0.17\,\mathcal{L}$.*

Proof. $x(x^2+12)$ on $x \equiv 5 \pmod{12}$ has no fixed prime divisor, so H gives infinitely many primes $p \equiv 5 \pmod{12}$ with $p^2 + 12$ prime. Here $\chi_{-12}(-p) = 1$. The target counts divisors $f \leq p$ of $p^2 + 12$ with $f \equiv -p \equiv 7 \pmod{12}$ (Lemma 2.1(iii)). The only one is $f = 1 \not\equiv 7$. □

Proposition 4.11 (CONDITIONAL on Dickson's conjecture for two linear forms; [RT2, Lemma 1.2.1])**.** *For every fixed (c, d) there are infinitely many primes p for which the Type II shell (c, d) is empty, although its mean is $\asymp \mathcal{L}/\varphi(4cd)$.*

Proof. The pair $X,\ 4c^2dX + 1$ has no fixed prime divisor, so Dickson's conjecture gives infinitely many primes p with $4c^2dp + 1$ prime. Its divisors 1 and $4c^2dp + 1$ are $\equiv 1 \not\equiv -1 \pmod{4cd}$. □

Under H, applied to finitely many polynomials at once, any fixed finite union of such targets is empty infinitely often. This is Theorem 3.2(iii) seen from the fixed-p side. EVIDENCE: all 3630 primes $p \equiv 5 \pmod{12}$ in $[10^6, 3 \cdot 10^6)$ with $p^2 + 12$ prime have the $(1, 3)$ target empty, and among all genus-allowed primes near 10^6, 13–89% of the individual targets (c, d) are empty [O123, §3, §5].

5. Quantitative gaps to known equidistribution

Everything in this section is an Assessment: a comparison of exponents, not a theorem about what any method can do.

5.1. **What is known.** Young [Yo, Thm. 2.1] proves a hybrid subconvexity bound which gives, for Heegner points of discriminant $-D$, an error of the shape $\|\phi\|_2 D^{5/12+\varepsilon} V^{1+\varepsilon}$ for a test function at Sobolev scale V. Consequently every fixed ball of radius $D^{-\delta}$ with $\delta < 1/24$ receives its expected share of points, and $\delta < 1/8$ under the Lindelöf hypothesis; see also Humphries [Hu], who also gives the genus and closed-geodesic versions. Young's Prop. 1.2 is a *single-discriminant existence theorem* proved spectrally: $b^2 \equiv -D \pmod{4a}$ has solutions with $a \asymp \sqrt{D}$, $|b| \asymp \sqrt{D}/V$ for every large odd fundamental D, in the range $V \ll D^{1/18-\varepsilon}$. It is the model of what a spectral pointwise ES statement would have to look like [LA]. Humphries and Radziwiłł[HR] reach much smaller scales, but only for almost all centres. The ES targets are specific regions attached to cusps, not random ones.

5.2. **What ES needs.**

- *Type I.* In the T_p-translated picture the discriminant is $D' = 4dp^2$ and the target tubes have width $\asymp 1/p \asymp |D'|^{-1/2}$ (Proposition 2.3; [O123, Prop. 3.2]): exactly the Planck scale. In the level-d picture $D = 4d$ is tiny and the large parameter enters only through the shape of the target. Proposition 4.6 shows that thickening to a reachable width $|D'|^{-\delta}$ produces averages over n in intervals of length $p^{1-2\delta}$.
- *Type II.* The shells of Proposition 2.5(c) have width $\asymp D^{-1/2}/c$ with $D = 4dp$. Young's method reaches shells of width $D^{-1/18}$ unconditionally and $D^{-1/6}$ under the Lindelöf hypothesis [O128, §4]. The gap is a factor $D^{4/9}$, resp. $D^{1/3}$, so what is needed goes beyond GLH.
- *Accuracy.* By Proposition 4.4 and [O123, Thm. 5.1], the count to be certified is $\le \psi\mathcal{L}^3$ for all but a proportion $O(1/\psi)$ of primes. Termwise use of GLH or of the Linnik–Selberg conjecture loses p^{ε}, which exceeds every power of $\mathcal{L}$; individual central values are not polylogarithmically bounded. GRH can enter with only polylogarithmic losses in averaged form (moments, effective Siegel). But a Cauchy–Schwarz estimate over the spectrum at Planck scale still loses a factor $\approx p$ against the main term $\approx \mathcal{L}$ [O123, §5].
- *Termwise GRH on a character expansion.* Expanding the $a = 1$ slice count into Dirichlet characters and bounding each twisted divisor sum by a GRH-type estimate gives error $\ge p^{1-o(1)}$ against a main term $\asymp \log^2 p$ [O127, Prop. 9.1] (PROVED bookkeeping, mechanism scope).

Assessment 5.1. No standard hypothesis (GRH, GLH/Lindelöf in any aspect, Ramanujan–Petersson, Selberg's eigenvalue conjecture, twisted Linnik–Selberg) is known to suffice, by the routes of this note, to prove ES, or the Type I or Type II part of ES, for all large primes. A sufficient statement would have to be at the Planck scale, of polylogarithmic accuracy, genus-aware (carrying $1 + \chi_{-4d}(-p)$, without which it is false at $n = p^2$) and aggregated over a growing set of levels and cusps. Such a statement is ES restated, not a standard conjecture. Its level window must also be anchored at the least quadratic non-residue n_p. For $p \equiv 1 \pmod 4$, Type I at level d needs $(-d/p) = -1$, so a window $d \le D(p)$ forces $n_p \le D(p)$; GRH gives $n_p \ll \mathcal{L}^2$ (Ankeny, Bach) and Burgess gives $n_p \ll p^{1/(4\sqrt{e})+\varepsilon}$ unconditionally.

6. Open problems: the escape hatches

By Theorem 3.2(iii), a route that survives must use a parameter set growing with p, integers that are not formal polynomials in p, non-abelian data, or a signed counting certificate. We list the concrete forms [RT2, §3]. The verdicts are Assessments. Each hatch is equally a hatch for *disproof*: a sterile prime, if one exists, would be a failure not explained by Brauer–Manin (Proposition 4.9), and none of the encodings above could detect it either.

Problem 1 (H-A: joint factorisation of a growing family)**.** Prove, at a single large prime p, a lower bound for $\sum_{\theta\in\Theta(p)} \#\{f \mid N_\theta(p) :\ f \bmod q_\theta \in R_\theta(p)\}$ for a family $\Theta(p)$ growing with p; for example $\sum_{a\le A,\, d\le D} \#\{f \mid p+4a^2d :\ f \equiv -1 \pmod{4ad}\}$ (row II$\{a,d\}$), or the Type I family p^2+4c^2d.

Divisor sums over the values of a *fixed* quadratic polynomial are classical [Ho], and roots of quadratic congruences are equidistributed on average over the modulus [DFI]. We know no result uniform in the constant term p in the needed range. More seriously, ES needs a *moving* modulus $q = 4ad$ tied to the variable. Once $q > N^{1/3}$, at most 11 divisors of N lie in one class [Le], so each θ contributes $O(1)$. The family is a sum of $\asymp \mathcal{L}^3$ events of size $O(1)$, with no large sub-family (§7). *Verdict:* open; no known technology. This is the main hatch, and any automorphic route (Kuznetsov over the level) lives here, with the polylogarithmic accuracy demanded by Proposition 4.4.

Problem 2 (H-B: parity-breaking for a divisor-in-moving-class event)**.** A slice is empty when $N_\theta(p)$ is prime (Propositions 4.10, 4.11). More generally it is empty when all prime factors of $N_\theta(p)$ lie in classes generating a subgroup that misses $R_\theta(p)$. Exclude, at a fixed p, the conspiracy in which every $N_\theta(p)$, $\theta \in \Theta(p)$, is bad in this sense.

"Bad" is a multiplicative-class condition of parity type, which lower-bound sieves cannot detect without bilinear input [FI, ch. 16]. Parity-breaking through bilinear sums exists for thin polynomial sequences averaged over their variables [FI98, HB, Ma]. At fixed p the variables θ are free to average over. The differences are the target (a divisor in a moving class, not primality) and the size of the family ($\asymp \mathcal{L}^3$ effective members). *Verdict:* open; the closest existing technology in spirit. The best-defined non-automorphic research target in this list is a "Friedlander–Iwaniec theorem for divisors in moving classes over the Elsholtz–Tao family at fixed p". The $O(1)$-per-block mass makes bilinear methods, which need long sums, hard to apply.

Problem 3 (H-D: non-abelian or higher-rank data)**.** Find an object carrying information not determined by congruences (Frobenius in non-abelian extensions, GL_3 or higher automorphic forms) with non-zero coefficient in an ES count.

By Proposition 2.9(iii), in the Type I CM torus the dihedral data have coefficient 0, so such data could only enter through the choice of θ, not by deciding a fibre. The varieties are toric or quadric (Lemma 3.1), so no natural higher-rank object is attached. *Verdict:* open, with no candidate object known.

Problem 4 (H-E: an exact identity with positive main term)**.** Find one expansion of an aggregated ES count, with a positive main term and cancellation between different θ that is not visible piecewise.

Termwise use of all standard hypotheses is excluded at the level of mechanism (§5); even the Linnik–Selberg conjecture for sums of Kloosterman sums (partial results: [ST]) loses p^{ε}. The search of [O131] found no structurally positive identity. Its exact content: $T_2^+(m^2) = 0$ for every m; for $cd \mid 6$ the Type II (c, d)-slice is an exact combination of binary theta coefficients at $4c^2dn + 1$; the Type II count is "Eisenstein even/odd parts minus a boundary -2 per level" [O131, Prop. 4.1]; and there is no linear relation between 56 ES count variants and 54 classical arithmetic functions on $n \le 2 \cdot 10^4$ (CERTIFIED, exact $\mathbb{Q}$-rank; the scope is that finite dictionary). In every case positivity is again "a specific integer has a divisor in a residue class". *Verdict:* open.

Remark 6.1 (H-F: integral Hasse principles, a cautionary example)**.** ES(p) asks for an integral point on an affine variety whose expected number of points is polylogarithmic. In the closest family where the integral Hasse principle has been settled, the Markoff-type surfaces, it holds for almost all parameters but fails for infinitely many [GS, Thm. 1.2]. In that setting "almost all" is what geometry delivered, not "all".

Remark 6.2 (non-escapes)**.** Anchoring the levels at the least non-residue n_p (H-C) fixes the genus condition but leaves H-A at the anchored levels. Taking Type I and Type II together (H-G) has the same $\mathcal{L}^3$ mass structure, and Theorem 3.2(iii) applies to unions of rows of both types [RT2, §3].

7. Numerical evidence

All statements here are EVIDENCE. The counts are exact enumerations, validated against brute force for $p \le 400$ and reproduced independently by internal reviewers, as indicated.

- $f_I(p) \ge 1$ and $f_{II}(p) \ge 1$ for every prime $5 \le p \le 10^8$, and the $a = 1$ Type I slice is non-empty for each of them [O123, §1], [O128, §10]. This was reproduced to 10^7 (Type I) and to 10^8 (Type II) by independent code.
- Dyadic minima of f_I grow with p and stay at 0.12–0.20 times the dyadic mean. The mean is $\approx 0.1\mathcal{L}^3$. For example $\min f_I = 105$ on $[2^{25}, 2^{26})$, at $p = 37117609$. All window minimisers from 2^9 on are Mordell-hard (in the six square classes modulo 840). The $a = 1$ slice alone has minimum 1–5 throughout.
- $\min f_{II}/\log^3 p \approx 0.012$ over $p \ge 10^5$, attained at $p = 2031121$; dyadic minima of f_{II} are $62, 74, 88, 94$ on the blocks $2^{23}, \dots, 2^{26}$ (the last one partial). The hardest primes lie in the square classes modulo 840; the 20 smallest normalised values are all $\equiv 1 \pmod{24}$ except one.
- Single fixed targets are often empty: the $d = 1$ Type II family (the single discriminant $-4p$) is empty for 148–198 primes per block at 2^{24}–2^{26}. Among primes $p \equiv 1 \pmod{24}$, $p < 10^5$, with $(-d/p) = -1$, Type I at the single level $d = 7, 11, 13$ fails for 42/612, 23/611, 88/608 primes, while $d = 5$ never fails. It is an "identity level": a classical polynomial covering [O126, §5.3]. Type II at level $d = 1$ fails for 73 of the 14183 primes $p \equiv 1 \pmod{24}$ below $1.5 \cdot 10^6$.
- The slice of [O127, Lemma 1.1] solves every Mordell-hard prime below 10^7, but its count drops to 1 at $p = 5101441$, where $f_I = 188$ and $f_{II} = 78$.

- Type II solutions near 10^6 sit at about one per dyadic (a, c, d) block [RT2, §3.1]: the mass is spread over $\asymp \mathcal{L}^3$ small events, with no large sub-family.

8. Summary

The automorphic and geometric objects naturally attached to (1.1) at a fixed prime (SL_2 matrices and Heegner points, T_p translates, shells, quaternion orders, CM tori, an isotropic ternary quadric, the Brauer class) are exact reformulations. For fixed p each of them reduces to genus or congruence data plus divisor events of specific integers at the Planck scale, with polylogarithmic expected counts (Theorem 3.2). Assessment: by these routes and with known technology, no standard hypothesis is known to suffice. What remains open is the joint control of a growing family of divisor events at one prime (H-A, H-B), or a genuinely new object (H-D, H-E). The Erdős–Straus conjecture remains open.

References

[ET] C. Elsholtz and T. Tao, Counting the number of solutions to the Erdős–Straus equation on unit fractions, *J. Aust. Math. Soc.* **94** (2013), 50–105; arXiv:1107.1010.

[BL] M. Bright and D. Loughran, Brauer–Manin obstruction for Erdős–Straus surfaces, *Bull. Lond. Math. Soc.* **52** (2020), 746–761; arXiv:1908.02526.

[Ya] K. Yamamoto, On the Diophantine equation $4/n = 1/x + 1/y + 1/z$, *Mem. Fac. Sci. Kyushu Univ. Ser. A* **19** (1965), 37–47. (Not accessed; cited through [BL, App. A].)

[Mo] L. J. Mordell, *Diophantine equations*, Academic Press, 1969, pp. 287–290.

[Sch] A. Schinzel, On sums of three unit fractions with polynomial denominators, *Funct. Approx. Comment. Math.* **28** (2000), 187–194.

[Pi] J. Pila, Density of integral and rational points on varieties, *Astérisque* **228** (1995), 183–187.

[Du] W. Duke, Hyperbolic distribution problems and half-integral weight Maass forms, *Invent. Math.* **92** (1988), 73–90.

[DSP] W. Duke and R. Schulze-Pillot, Representation of integers by positive ternary quadratic forms and equidistribution of lattice points on ellipsoids, *Invent. Math.* **99** (1990), 49–57.

[Yo] M. P. Young, Weyl-type hybrid subconvexity bounds for twisted L-functions and Heegner points on shrinking sets, *J. Eur. Math. Soc.* **19** (2017), 1545–1576; arXiv:1405.5457.

[Hu] P. Humphries, Equidistribution in shrinking sets and L^4-norm bounds for automorphic forms, *Math. Ann.* **371** (2018), 1497–1543.

[HR] P. Humphries and M. Radziwiłł, Optimal small scale equidistribution of lattice points on the sphere, Heegner points, and closed geodesics, *Comm. Pure Appl. Math.* **75** (2022), 1936–1996; arXiv:1910.01360.

[GZ] B. H. Gross and D. B. Zagier, On singular moduli, *J. Reine Angew. Math.* **355** (1985), 191–220.

[GS] A. Ghosh and P. Sarnak, Integral points on Markoff type cubic surfaces, *Invent. Math.* **229** (2022), 689–749; arXiv:1706.06712.

[Le] H. W. Lenstra, Jr., Divisors in residue classes, *Math. Comp.* **42** (1984), 331–340.

[Ho] C. Hooley, On the number of divisors of a quadratic polynomial, *Acta Math.* **110** (1963), 97–114.

[DFI] W. Duke, J. Friedlander and H. Iwaniec, Equidistribution of roots of a quadratic congruence to prime moduli, *Ann. of Math.* **141** (1995), 423–441.

[FI] J. Friedlander and H. Iwaniec, *Opera de cribro*, AMS Colloquium Publications **57**, 2010.

[FI98] J. Friedlander and H. Iwaniec, The polynomial $X^2 + Y^4$ captures its primes, *Ann. of Math.* **148** (1998), 945–1040.

[HB] D. R. Heath-Brown, Primes represented by $x^3 + 2y^3$, *Acta Math.* **186** (2001), 1–84.

[Ma] J. Maynard, Primes represented by incomplete norm forms, *Forum Math. Pi* **8** (2020), e3.

[ST] P. Sarnak and J. Tsimerman, On Linnik and Selberg's conjecture about sums of Kloosterman sums, in *Algebra, Arithmetic, and Geometry*, Progr. Math. **270**, Birkhäuser, 2009, 619–635.

[PO] *Pointwise obstructions for the Erdős–Straus seed-component programme*, internal note, `paper/pointwise-obstruction.tex`.

[PS] *Formal genericity and pointwise* Ω*-results*, internal note, `POINTWISE_SIZE.md` (Theorems M and C), with internal review `reviews/pointwise-size-step1-review.md`.
[SE] *Large Erdős–Straus witnesses II*, internal note, `paper/es-subexp-note.tex`.
[TH] *The Elsholtz–Tao Type I sum is* $O(N\log^2 N)$, internal note, `paper/es-typei-heegner-note.tex` (the bound (D)32b).
[O123] *Flip the average: fixed-p spectral forms of Type I*, internal note, `POINTWISE_SPECTRAL.md`, review `reviews/pointwise-spectral-review.md`.
[O124] *Embedding ES into representations by quadratic forms*, internal note, `POINTWISE_TERNARY.md`, review `reviews/pointwise-ternary-review.md`.
[O126] *Arithmetic geometry of the Erdős–Straus surfaces*, internal note, `GEOMETRY_ES.md`, review `reviews/geometry-es-review.md`.
[O127] *ES for large primes conditional on a standard conjecture?*, internal note, `CONDITIONAL_ES.md`, review `reviews/conditional-es-review.md`.
[O128] *Automorphic structure of Type II solutions for fixed p*, internal note, `POINTWISE_TYPEII_SPECTRAL.md`, reviews `reviews/pointwise-typeii-spectral-review.md`, `-review-r2.md`.
[O130] *Quaternionic / supersingular reformulation*, internal note, `QUATERNION_ES.md`, review `reviews/quaternion-es-review.md`.
[O131] *Exact-formula search*, internal note, `EXACT_FORMULA.md`, review `reviews/exact-formula-review.md`.
[RT1] *Red team for the automorphic/geometric attack*, internal report, `reviews/redteam-automorphic.md`.
[RT2] *Red team 2 and synthesis: the divisor-class reduction*, internal report, `reviews/redteam-automorphic-2.md`.
[LA] *Spectral and geometric tools for pointwise representation (literature survey)*, internal file, `LITERATURE_AUTOMORPHIC.md`.